belinus®

Module · Gaia Smart
v01 · 2026-01
MODEL · belinus Gaia Smart

● NEW · PRODUCT LAUNCH 2027

BACK-CONTACT HBC MODULE · INTEGRATED OPTIMIZER · 490–505 WP

Gaia Smart

Unlock the power of nature.

#495Wp #IntegratedOptimizer #SingleSKU #BackContact #GlassGlass #MadeInBelgium

01 **Integrated optimizer — single SKU.** Module-level MPPT and monitoring built into every panel; no external optimizer to mount or wire.

02 **Module-level safety.** SunSpec-compliant rapid shutdown and per-module shutdown on fault.

03 **Up to 24.7 % efficiency.** < 1 % first-year degradation, 0.35 % / yr from year 2.

04 **Perfect black, glass-glass.** No gridlines, IP68 junction box, butyl-sealed.

05 **Made in Belgium — full product.** Module and integrated optimizer assembled in Belgium.

06 **Bifacial 0BB design.** Rear-side power gain up to 25 %; 6000 / 3600 Pa load.

● GLOBAL LIGHTHOUSE NETWORK · WEF AWARDED FACTORY

● MADE IN BELGIUM — FULL PRODUCT

35 yr	35 yr	35 yr
PRODUCT WARRANTY	PERFORMANCE WARRANTY	SERVICE WARRANTY

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TECHNICAL SPECIFICATIONS

Gaia Smart · Back-Contact Module with Integrated Optimizer

MAX POWER	MAX EFFICIENCY	POWER DENSITY	OPTIMIZER
505 Wp	24.7 %	247 W/m²	Integrated

MECHANICAL

Cells	108 (6 × 18) back-contact
Dimensions	1800 × 1134 × 30 mm
Weight	23.5 kg
Frame	Black anodized aluminium
Glass	Dual glass 2.0 + 1.6 mm
Junction box	IP68
Mechanical load	6000 Pa / 3600 Pa
Fire rating	IEC Class A

ELECTRICAL (STC)

Max power (Pmax)	490 – 505 Wp
Open-circuit voltage (Voc)	41.25 – 41.55 V
Short-circuit current (Isc)	15.04 – 15.37 A
Voltage at Pmax (Vmp)	34.07 – 34.37 V
Current at Pmax (Imp)	14.38 – 14.69 A
Module efficiency	24.0 – 24.7 %
Power tolerance	0 ~ +3 %

MODULE

Cell technology	HBC back-contact (0BB)
Construction	Glass-glass, perfect black
Appearance	No gridlines
Bifacial gain	up to 25 %
First-year degradation	< 1 %
Annual degradation	0.35 % (yr 2–30)

TEMPERATURE & OPERATING

Temp. coeff. (Pmax / Voc / Isc)	−0.24 / −0.21 / +0.05 %/°C
NOCT	45 ± 2 °C
Operational temperature	−40 °C to +85 °C
Max system voltage	DC 1500 V (IEC)
Max series fuse	30 A
Protection class	Class II

INTEGRATED OPTIMIZER

Type	Module-level · single SKU
Functions	Per-module MPPT · monitoring
Safety	SunSpec rapid shutdown
Communication	Power-line to gateway

CERTIFICATION

Standards	IEC 61215 · IEC 61730
Management	ISO 9001 · 14001 · 45001 · 62941
Origin	Made in Belgium
Factory	Global Lighthouse Network (WEF)

WARRANTY

Product	35 yr
Performance	35 yr
Service	35 yr

STRATEGIC OEM PARTNER LONGi